AI WILDLIFE COLLISION PREVENTION AND ROAD SAFETY ALERT SYSTEM

A MINI PROJECT REPORT

Submitted by

MONISHA V (612924101112)

MAVISHA T (612924101104)

POOJASRI G (612924103124)

KANISHKA S (612924103067)

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COMPUTER SCIENCE AND ENGINEERING

&

ELECTRONICS AND COMMUNICATION ENGINEERING





VIVEKANANDHA

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VIVEKANANDHA COLLEGE OF ENGINEERING FOR WOMEN

BONAFIDE CERTIFICATE

Certified that this project report **"AI WILDLIFE COLLISION PREVENTION AND ROAD SAFETY ALERT SYSTEM"** is the bonafide work of **"MONISHA V,MAVISHA T,POOJASRI G,KANISHKA S"** who carried out the mini project work under my supervision.

1.

2.

SIGNATURE

Dr.D.RADHIKA

Dr. P.T.KALAIVAANI

HEAD OF THE DEPARTMENT

Associate Professor & Head

Department of Computer Science & Engineering

Professor & Head

Department of Electronics & Communication Engineering

Vivekanandha College of Engineering for Women(Autonomous), Namakkal.

1.

2.

SIGNATURE

Mr.M.RAMESH

Dr.S.SASIKANTH

SUPERVISOR

Assistant Professor

Department of Computer Science & Engineering

Assistant Professor

Department of Electronics & Communication Engineering

Vivekanandha College of Engineering for Women(Autonomous), Namakkal.

Submitted to the Viva voce Examination held on ……

INTERNAL EXAMINER

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Institute Vision:

To impart value-based education in Engineering and Technology to empower young women to meet the societal exigency with a global outlook.

Institute Mission:

- To provide holistic education through innovative teaching-learning practices
- To instill self confidence among rural students by supplementing with co-curricular and extra-curricular activities
- To inculcate the spirit of innovation through training, research and development
- To provide industrial exposure to meet the global challenges
- To create an environment for continual progress through lifelong learning

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- To promote lifelong learning, Leadership and social responsibility for societal development.

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PEO 3

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Program Outcomes:

Program Specific Outcomes:

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- **PSO2:** Design, verify and validate electronic functional elements for numerous applications including signal processing, communications, computer networks and VLSI.
- **PSO3:** Collaborate with peer engineers and multidisciplinary teams to develop innovative, cost- effective, and appropriate solutions for various societal challenges.

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- To promote lifelong learning, Leadership and social responsibility for societal development.

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Program Outcomes:

- **PO1: Engineering Knowledge:** Apply knowledge of mathematics, science, and engineering fundamentals.
- **PO 2. Problem Analysis:** Identify, formulate, and analyze complex engineering problems.
- **PO 3. Design/Development of Solutions:** Design solutions for complex problems and system components.
- **PO 4. Conduct Investigations of Complex Problems:** Use research-based knowledge to conduct investigations.
- **PO 5. Engineering Tool Usage:** Select and apply modern tools and techniques.
- **PO 6. The Engineer and the World:** Understand professional responsibilities to society and sustainable development.
- **PO 7. Ethics and Inclusive Behavior:** Apply ethical principles, commit to professional ethics, and embrace diversity.
- **PO 8. Individual and Collaborative Team Work:** Function effectively as an individual and in multidisciplinary teams.
- **PO 9. Communication:** Communicate effectively on complex activities.

- **PO 10. Project Management and Finance:** Apply engineering and management principles to manage projects.
- **PO 11. Life-Long Learning:** Recognize the need for independent and life-long learning and adaptability

Program Specific Outcomes:

- **PSO1:** Comprehend the basic concepts of electronics and communication and apply in the day to day life to design and execute complete engineering systems.
- **PSO2:** Design, verify and validate electronic functional elements for numerous applications including signal processing, communications, computer networks and VLSI.
- **PSO3:** Collaborate with peer engineers and multidisciplinary teams to develop innovative, cost- effective, and appropriate solutions for various societal challenges.

Course Outcome mapping with PO's and PSO's

Course code & Name:	U23CS517& Mini Project I U23EC507& Mini Project I	**Programme & Branch:**	B.E &Computer science& Engineering B.E & Electronics &Communication Engineering
Faculty Name:	Mr.M.Ramesh Dr.S.Sasikanth	**Regulation:**	2023
Academic Year	2026-2027	**Semester:**	V
Title of the Project:	**AI WILDLIFE COLLISION PREVENTION AND ROAD SAFETY ALERT SYSTEM**		

COURSE OUTCOMES (COs):

Upon completion of the course, students will be able to:

CO#	**Course Outcomes**	**Bloom's Taxonomy Level**
CO1	Identify the fundamental concepts of Artificial Intelligence, computer vision, and wildlife object detection.	Remember
CO2	Explain the working principles of YOLO and OpenCV for detecting wildlife in video frames.	Understand
CO3	Implement an AI-based wildlife detection system using Python, OpenCV, and the YOLO model.	Apply
CO4	Analyze detection confidence, estimated distance, and collision risk levels to identify potential road hazards.	Analyze
CO5	Evaluate the system's performance in detecting wildlife and assessing collision risks, and recommend improvements for road safety.	Evaluate

CO – PO / PSO Mapping

(3/2/1 indicates strength of correlation) 3 – Strong, 2 – Medium, 1 – Weak

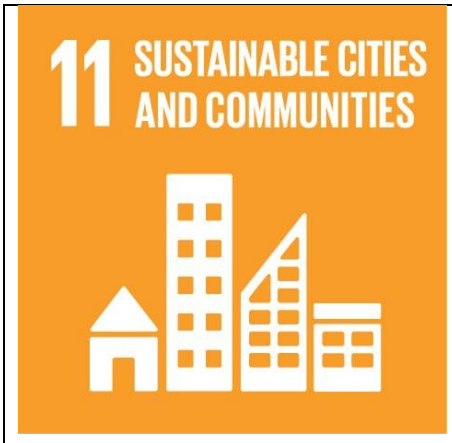
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PS03
CO1	2	1			1	1						2	1	
CO2	2	2	1		2	1						2	2	1
CO3	2	2	3	2	3	1	1		1			3	2	3
CO4	2	3	2	3	2	2	2	1	1			2	3	3
CO5	2	3	2	3	2	2	2	1	1			2	2	3

Relevance to PO's and PSO's:

PO1	**Engineering Knowledge** Apply engineering principles, mathematics, and technical knowledge to develop the proposed system.
PO2	**Problem Analysis** Identify and analyze wildlife–vehicle collision problems using appropriate engineering concepts.
PO3	**Design/Development of Solutions** Design an AI-based system to detect wildlife and assess potential collision risks.
PO4	**Conduct Investigations of Complex Problems** Test the system using video inputs and analyze detection results and limitations.
PO5	**Modern Tool Usage** Use Python, YOLO, and OpenCV for object detection and video processing.

PO6	**The Engineer and Society** Understand how AI-based wildlife detection can contribute to road safety and public welfare.
PO7	**Environment and Sustainability** Support wildlife protection and promote safer coexistence between humans and animals.
PO8	**Ethics** Use AI technology responsibly and report system capabilities and limitations honestly.
PO9	**Individual and Teamwork** Coordinate project activities and collaborate effectively to achieve project objectives.
PO10	**Communication** Present the project methodology, results, and conclusions clearly through reports and presentations.
PO11	**Project Management and Finance** Plan project activities, manage available resources, and consider implementation costs.
PSO1	**Professional Skills** Apply computer science and engineering knowledge to develop AI-based solutions for wildlife detection and road-safety assessment.
PSO2	**Problem-Solving Skills** Use programming, computer vision, and AI techniques to identify wildlife and assess potential collision risks.

Mapped SDGs:

	• Improves road safety by identifying wildlife near roadways and assessing potential collision risks. • Supports intelligent transportation through proposed warning indicators, audible alerts, and GPS-based notifications.
	• Protects wildlife and biodiversity by detecting animals near roads using AI-based object detection. • Aims to reduce wildlife–vehicle collisions and promote the safe coexistence of humans and animals.

CO-PO/PSO Mapping:

Project Title:	**AI WILDLIFE COLLISION PREVENTION AND ROAD SAFETY ALERT SYSTEM**													
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PS03
CO1	✓	✓			✓							✓	✓	
CO2	✓	✓	✓		✓							✓	✓	✓
CO3	✓	✓	✓	✓	✓							✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓	✓		✓	✓		✓	✓	✓
CO5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓

ABSTRACT

Wildlife-vehicle collisions are a major concern on roads passing through forest areas, affecting both animal conservation and human safety. Traditional warning signs may not provide timely alerts when animals suddenly cross roads, especially during night-time and low-visibility conditions. The proposed AI Wildlife Collision Prevention and Road Safety Alert System aims to detect animals near roads and estimate potential collision risks using computer vision and artificial intelligence.

The system uses Python, OpenCV and the YOLO object detection model to process video frames and identify animals. Based on the detected object, its confidence score and estimated distance, the system classifies the collision risk into different levels. A software-based prototype is being developed to demonstrate animal detection and risk assessment. Future hardware integration using an ESP32-CAM, LEDs, a buzzer and GPS can provide visual, acoustic and location-based alerts to drivers. The proposed system aims to improve road safety, reduce wildlife mortality and support wildlife conservation through intelligent monitoring.

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LIST OF SYMBOLS AND ABBREVIATIONS

S. NO.	SYMBOL / ABBREVIATION	DESCRIPTION
1	AI	Artificial Intelligence
2	CV	Computer Vision
3	YOLO	You Only Look Once
4	OpenCV	Open Source Computer Vision Library
5	ML	Machine Learning
6	FPS	Frames Per Second
7	CPU	Central Processing Unit
8	GPU	Graphics Processing Unit
9	IDE	Integrated Development Environment
10	Python	Programming Language Used for Implementation
11	IoT	Internet of Things
12	GPS	Global Positioning System
13	ESP32-CAM	ESP32-Based Camera Module
14	LED	Light-Emitting Diode
15	m	Metre
16	%	Percentage
17	CO	Confidence Score

18	HIGH	High Collision Risk Level
19	MEDIUM	Medium Collision Risk Level
20	LOW	Low Collision Risk Level

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Wildlife–vehicle collisions are a major concern on roads passing through forests and wildlife habitats. These accidents can cause animal fatalities, vehicle damage, and human injuries.

Artificial Intelligence (AI) and computer vision can help detect animals in video input and identify potential road hazards. This project uses Python, OpenCV, and the YOLO object detection model to detect objects, display confidence scores, estimate distance, and classify collision risk.

1.2 WILDLIFE–VEHICLE COLLISIONS

Wildlife–vehicle collisions occur when animals cross roads unexpectedly. Poor visibility, high vehicle speed, and delayed driver reactions can increase the risk of accidents. AI-based detection can support road safety by identifying objects in video frames and indicating potential hazards.

1.3 NEED FOR AI-BASED WILDLIFE DETECTION

Manual observation may not always provide sufficient warning when animals suddenly enter roads. AI-based object detection can automatically identify objects in video frames. Using YOLO and OpenCV, the proposed system detects objects and displays information that supports collision risk assessment.

1.4 OVERVIEW OF THE PROPOSED SYSTEM

The proposed system processes video input to detect objects and assess potential collision risks. Python, OpenCV, and YOLO are used for frame processing and object detection.

The main stages are:

1. Video input acquisition.
2. Frame processing using OpenCV.
3. Object detection using YOLO.
4. Display of object labels and confidence scores.
5. Estimated distance calculation.
6. Collision risk classification.

The current prototype is software-based. GPS notifications and hardware warning indicators are planned as future enhancements.

1.5 CHALLENGES IN WILDLIFE DETECTION

The major challenges include:

- **Lighting conditions:** Darkness and shadows affect visibility.
- **Complex backgrounds:** Vegetation can obscure animals.
- **Detection accuracy:** Objects may be missed or incorrectly classified.
- **Distance estimation:** Accurate measurement requires calibration or depth information.

- **Real-time processing:** Video frames must be processed efficiently.
- **Risk assessment:** Reliable classification may require distance, movement, and vehicle speed information.

1.6 PROBLEM IDENTIFICATION

Roads near wildlife habitats present a risk of collisions between animals and vehicles. Conventional warning signs cannot dynamically detect animals on the road.

This project addresses the problem by developing a software prototype that detects objects in video frames and displays confidence scores, estimated distances, and collision risk categories.

1.7 NOVELTY

The proposed system combines YOLO-based object detection with collision risk assessment. It displays object labels, confidence scores, estimated distances, and risk categories in a single software prototype.

The modular design also provides a foundation for future GPS notifications, warning LEDs, audible alerts, and hardware integration.

1.8 OBJECTIVES

The main objectives are:

1. To develop a video-processing system using Python and OpenCV.
2. To detect objects using the YOLO model.
3. To display object labels and confidence scores.
4. To estimate distance and classify collision risk using the implemented logic.
5. To test and analyse the detection outputs.

6. To identify improvements for a future road safety alert system.

1.9 SCOPE OF THE PRESENT WORK

The project focuses on software-based object detection and collision risk assessment using Python, OpenCV, and YOLO. The prototype processes video frames and displays detection results, estimated distances, and risk categories.

Physical distance measurement, GPS notifications, and hardware-based alerts are outside the current implementation. Further testing with wildlife datasets and real-world road footage is required to evaluate performance.

1.10 ORGANISATION OF THE REPORT

This report consists of five chapters.

Chapter 1 – Introduction: Describes the project background, problem, objectives, scope, and proposed system.

Chapter 2 – Literature Review: Reviews existing wildlife detection techniques and identifies the research gap.

Chapter 3 – Materials and Methods: Explains the software tools, system design, methodology, and risk assessment process.

Chapter 4 – Results and Discussion: Presents detection outputs, confidence scores, risk classifications, and limitations.

Chapter 5 – Conclusion: Summarises the work and discusses future enhancements.

CHAPTER 2

LITERATURE REVIEW

2.1 REVIEW OF EXISTING WORK

Wildlife detection and road safety have gained attention due to the increasing risk of animal–vehicle collisions. Traditional methods, such as warning signs and roadside barriers, help reduce accidents but cannot automatically detect animals entering roads.

Recent developments in Artificial Intelligence (AI) and computer vision have enabled automated object detection using images and videos. Deep learning models can identify objects and provide their locations and confidence scores. These technologies offer potential applications in wildlife monitoring and road safety systems.

2.2 AI-BASED WILDLIFE DETECTION TECHNIQUES

AI-based wildlife detection uses computer vision and deep learning to identify animals from images or video frames. Common approaches include image classification, object detection, and image segmentation.

YOLO (You Only Look Once) is an object detection model that identifies objects and predicts their locations in a frame. It is suitable for applications requiring efficient detection.

OpenCV is a computer vision library used to capture video, process frames, and display detection results. Integrating OpenCV with YOLO enables video-based object detection.

In this project, YOLO is used to detect objects, while OpenCV handles video processing and result display. Confidence scores and implemented risk-assessment logic are used to present potential collision hazards.

2.3 RESEARCH GAP

Existing road safety methods often depend on fixed warning signs or manual observation. These methods do not dynamically identify animals in video input.

AI-based object detection provides an opportunity to identify objects automatically, but practical systems must also consider detection accuracy, environmental conditions, distance estimation, and timely alerts.

The present project addresses this area by combining video-based object detection with confidence score display, approximate distance estimation, and collision risk classification. The current prototype is software-based and requires further validation under real-world conditions.

2.4 SUMMARY OF THE LITERATURE

The literature review highlights the potential of AI and computer vision for wildlife detection and road safety applications. YOLO supports object detection, while OpenCV enables video capture and frame processing.

Combining these technologies provides a foundation for detecting objects and assessing potential collision risks. The proposed prototype demonstrates this approach through detection outputs and risk categories. Future work can improve reliability through suitable wildlife datasets, calibrated distance measurement, and integration with GPS and hardware-based warning systems.

CHAPTER 3

MATERIALS AND METHODS

3.1 METHODOLOGY

The project follows a software-based approach to detect objects in video frames and assess potential collision risks. Python, OpenCV, and the YOLO object detection model are used for implementation.

3.1.1 OVERALL APPROACH

The system takes video input, processes individual frames, detects objects using YOLO, and displays object labels and confidence scores. The implemented logic estimates distance and classifies the potential collision risk.

3.1.2 STAGES OF THE PROJECT

The project consists of the following stages:

1. Collect video input.
2. Process video frames using OpenCV.
3. Detect objects using YOLO.
4. Display object labels and confidence scores.
5. Estimate distance using the implemented logic.
6. Classify and display collision risk levels.
7. The major stages involved in the project are listed in Table 3.1.

Table 3.1: Stages Involved in the Project Methodology

Stage	Description
1	Video input acquisition
2	Frame processing using OpenCV

3 Object detection using YOLO

4 Confidence score display

5 Estimated distance calculation

6 Collision risk classification

7 Output display

3.1.3 BLOCK DIAGRAM

The overall workflow of the proposed system is:



Figure 3.1: Overall Methodology of the Proposed System

3.2 MATERIALS AND SOFTWARE TOOLS USED

The current prototype uses software tools for video processing, object detection, and output display. Hardware components are planned for future development.

3.2.1 PYTHON PROGRAMMING LANGUAGE

Python is used to develop the application and integrate the object detection model with video processing and risk assessment.

3.2.2 YOLO OBJECT DETECTION MODEL

YOLO (You Only Look Once) is used to detect objects in video frames. It provides object labels, bounding boxes, and confidence scores.

3.2.3 OPENCV LIBRARY

OpenCV is used to capture video frames, process images, and display detection results.

3.2.4 VIDEO INPUT AND FRAME PROCESSING

The video input is processed frame by frame. Each frame is passed to the detection model for identifying objects in the scene.

3.2.5 WILDLIFE DETECTION MODULE

The detection module applies YOLO to identify objects supported by the model. Detected objects are displayed with their labels and bounding boxes. Wildlife detection performance depends on the model and input conditions.

3.2.6 CONFIDENCE SCORE ANALYSIS

The confidence score indicates the model's confidence in a predicted detection. It is displayed alongside the detected object to help interpret the output. A higher score does not guarantee that the prediction is correct.

3.2.7 DISTANCE ESTIMATION MODULE

The prototype displays an estimated distance using its implemented logic. This value is approximate and should not be treated as a validated physical distance measurement without suitable calibration.

3.2.8 COLLISION RISK CLASSIFICATION

The system assigns a risk category based on its implemented distance and risk-assessment logic. The displayed categories are HIGH, MEDIUM, and LOW. Further calibration and testing are required to establish reliable risk thresholds.

3.2.9 ROAD SAFETY ALERT MODULE

The current prototype displays detection and risk information on the screen. GPS notifications, warning LEDs, and audible alerts are proposed for future hardware integration.

3.2.10 DEVELOPMENT ENVIRONMENT

The prototype is developed on a Windows computer using Python, OpenCV, and the Ultralytics YOLO framework.

3.3 SYSTEM DESIGN

The system consists of video input, frame processing, object detection, confidence score display, distance estimation, risk classification, and output display modules. The modular design allows individual components to be improved or extended in future versions.

3.4 WILDLIFE DETECTION AND RISK ASSESSMENT

Video frames are analysed by the YOLO model to identify objects. The application displays the detection class and confidence score. Its implemented

logic then estimates distance and assigns a collision risk category. The results depend on detection quality and the accuracy of the estimation method.

3.5 WORKING PRINCIPLE

The system captures video input and processes it using OpenCV. Each frame is analysed by YOLO to detect objects. The detected labels and confidence scores are displayed, followed by the estimated distance and risk category. The output helps demonstrate how object detection can support wildlife-related road hazard assessment.

3.6 DATA PROCESSING AND ANALYSIS

The system processes video frames and displays the corresponding detection results. The output can be analysed using object labels, confidence scores, estimated distances, and risk categories. Testing with different videos is necessary to evaluate detection behaviour and identify limitations.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Video Input and Frame Processing

The developed system accepts video input and processes individual frames using OpenCV. Each frame is passed to the YOLO object detection model for identifying objects present in the video. The program processes the frames sequentially and displays the detection results during execution.

Figure 4.1: Video Input During System Execution

Insert a screenshot of your program running with the input video.

4.2 Wildlife/Object Detection Output

The YOLO model identifies supported object classes in the video frames and displays bounding boxes around detected objects. Each detection is associated with an object label and a confidence score. This output demonstrates the use of AI-based object detection for identifying objects that may be relevant to road safety.

Figure 4.2: Wildlife/Object Detection Output

Insert an actual screenshot showing the detected object and its bounding box.

4.3 Detection Confidence and Estimated Distance Analysis

The system displays a confidence score for each detected object. The score indicates the model's confidence in its predicted class and does not guarantee that the prediction is correct.

The program also displays an estimated distance for detected objects. This value is generated using the current implementation and should be treated as approximate because it has not been established as a calibrated real-world distance measurement.

Table 4.1: Sample Wildlife/Object Detection Results

S. No.	Detected Object	Confidence Score	Estimated Distance	Risk Level
1	Person	Approximately 48%	20 m	HIGH

2	Zebra	Approximately 42.52%	20 m	HIGH
3	Zebra	Approximately 57.46%	20 m	HIGH
4	Zebra	Approximately 44.2%	40 m	MEDIUM

Note: These values are examples from previously observed program outputs. Verify them against your actual output before final submission.

Figure 4.3: Detection Output with Confidence Scores

Insert an actual screenshot displaying object labels and confidence scores.

4.4 Collision Risk Classification

The system classifies detected objects into HIGH, MEDIUM, or LOW risk categories based on the estimated distance and the thresholds defined in the program. This feature provides a simple visual indication of potential risk.

The classification is dependent on the implemented rules. Therefore, it should be considered a preliminary risk-assessment output rather than a validated prediction of an actual collision.

Table 4.2: Detection Confidence and Estimated Distance Analysis

S. No.	Parameter	Discussion
1	Object Detection	The model identifies objects in the processed video frames.
2	Confidence Score	Indicates the model's confidence in each predicted class.

3	Estimated Distance	Provides an approximate value generated by the implemented logic.
4	Risk Classification	Assigns a risk category using the programmed thresholds.
5	Output Display	Presents the detection and risk information to the user.

Figure 4.4: Estimated Distance and Collision Risk Output

Insert an actual screenshot showing the estimated distance and risk category.

4.5 System Testing and Performance Analysis

The software prototype can be evaluated by checking video loading, frame processing, object detection, confidence score display, distance estimation, and risk classification. Each function should be verified by running the program with suitable input videos.

Table 4.3: System Testing and Performance Analysis

S. No.	Test Case	Expected Result	Observation
1	Load video input	Video opens successfully.	Record the actual result.
2	Process video frames	Frames are read and processed.	Record the actual result.
3	Detect objects	Supported objects are identified.	Record the actual result.
4	Display confidence scores	Scores appear with detections.	Record the actual result.

5	Estimate distance	An approximate distance is displayed.	Record the actual result.
6	Classify risk	A risk category is displayed.	Record the actual result.

The system's performance may vary depending on video quality, lighting conditions, object visibility, and processing capability. Actual accuracy, processing speed, and reliability should be reported only when measured during testing.

4.6 Applications, Benefits and Limitations

The proposed system demonstrates a possible application of AI-based object detection in wildlife-related road-safety monitoring. Its current software implementation provides a foundation for further development.

Table 4.4: Applications, Benefits and Limitations of the Proposed System

Category	Description
Application	Preliminary wildlife detection using road or wildlife video.
Application	Visual assessment of potential roadside risks.
Benefit	Automates object detection in video frames.
Benefit	Displays detection labels, confidence scores, and risk categories.
Limitation	Estimated distance may not correspond to actual physical distance.
Limitation	Detection performance may vary with lighting and video quality.
Limitation	Real-world collision prevention has not yet been validated.

Future Development	GPS notifications, ESP32-CAM integration, LEDs, and buzzer alerts are planned enhancements.

4.7 Discussion

The results demonstrate the basic functionality of the software prototype in processing video frames, detecting objects, displaying confidence scores, and assigning risk categories using programmed logic. These features provide a starting point for an AI-assisted wildlife road-safety system.

However, the current implementation has limitations. The distance estimates require further validation, and the risk categories depend on the thresholds used in the program. Additional testing with wildlife videos captured under different environmental conditions is necessary to assess detection reliability.

Hardware-based alerts and GPS notifications could be incorporated in future development to extend the system beyond software-based visual monitoring.

Overall, the prototype demonstrates the feasibility of combining object detection with rule-based risk assessment, while further calibration and real-world testing are required before practical deployment.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENTS

5.1 Conclusion

The project titled **"AI Wildlife Collision Prevention and Road Safety Alert System"** focuses on detecting animals and other objects in video frames using Artificial Intelligence. The system uses Python, YOLO, and OpenCV to process video input, identify objects, display detection confidence scores, estimate approximate distances, and classify collision risk into HIGH, MEDIUM, and LOW categories.

The developed software prototype demonstrates how AI-based object detection can support preliminary wildlife-related road-safety assessment. It provides visual information about detected objects and their estimated risk levels, which can help in identifying potentially dangerous situations on roads near wildlife habitats.

However, the current prototype has limitations, particularly in distance estimation and real-world validation. The system has been developed and tested as a software-based prototype, while hardware-based alerts and GPS notifications remain future enhancements.

5.2 Future Enhancements

The proposed system can be improved in the following ways:

1. **ESP32-CAM Integration:** A camera module can be integrated to capture live road images for wildlife detection.
2. **GPS-Based Location Tracking:** GPS can be added to identify the approximate location of detected wildlife and support location-based alerts.
3. **LED Warning Indicators:** Red, yellow, and green LEDs can indicate different collision-risk levels.
4. **Buzzer Alert System:** A buzzer can provide an audible warning when a high-risk situation is detected.

5. **Improved Distance Estimation:** Camera calibration, depth estimation, or additional sensors can improve the accuracy of distance measurements.
6. **Real-Time Video Processing:** The system can be optimized for faster processing and continuous detection from live video.
7. **Model Improvement:** Training or fine-tuning a detection model with suitable wildlife images can improve detection performance in forest and roadside environments.
8. **Field Testing:** Testing under different lighting, weather, and road conditions can help evaluate the reliability of the system.

5.3 Final Outcome

The project establishes a software-based foundation for AI-assisted wildlife detection and road-safety risk assessment. With further improvements in distance estimation, real-time processing, hardware integration, and field testing, the system could be developed into a more practical wildlife collision prevention solution.

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